CALCULUS · UNIT 1: PRECALCULUS REVIEW

Phone Battery Lab

Review function families by modeling a phone's battery, charging, and app data.

CP2

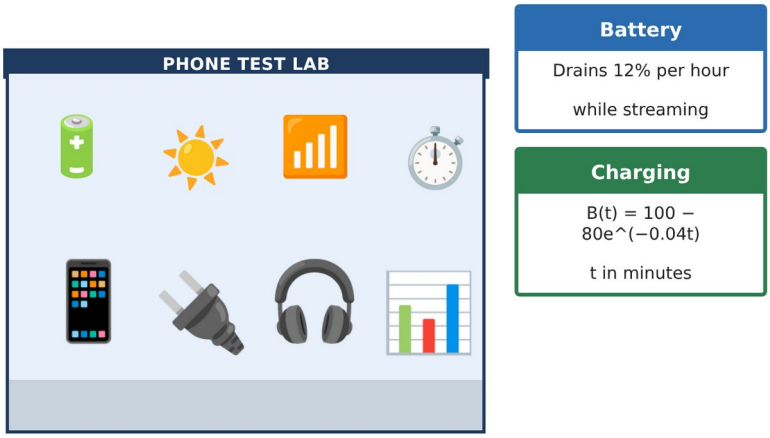
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A phone company wants a student lab to test its new phone. The battery drains while streaming, charges quickly at first and then slowly, the screen brightness follows the sunlight, and the data plan has a price that changes after a limit.

You will use the function families you need for calculus to model the phone and answer the company's questions.



Learning Goals

- I can write and use linear and exponential models.
- I can evaluate piecewise functions and compositions.
- I can use logarithms to solve for time.

Project Timeline

When	What you will do
Day 1	Battery Models
Day 2	Plans & Conversions
Day 3	Charging & Brightness
Final	Battery Lab Report + reflection

Part 1: Battery Models

Day 1 · Use the Lab Data Cards. t is in hours.

Lab Data Cards

Streaming	Gaming	Charging
Starts at 95%	Loses 18% of the	B(t) = 100 − 80e^(−0.04t)
loses 12% per hour	remaining charge per hour	

Data Plan	Brightness	Temperature
$30 up to 15 GB	Max 900 nits at 1 p.m.	F = 1.8C + 32
+$4 per GB over	Min 100 nits at 1 a.m.	

1. Write a linear model for streaming: S(t) = 95 − 12t. When is the battery at 20%?

2. Write an exponential model for gaming starting at 95%.

   Starter: $G(t) = 95(1 - ___)^t$

3. Find G(3) and S(3). Which drains faster in the first 3 hours?

4. What is the domain of S in this context?

Part 2: Plans & Conversions

Day 2 · Data plan: $30 up to 15 GB, then $4 per GB over 15.

5. Write the cost as a piecewise function P(g).

6. Find P(12) and P(20).

7. The phone's temperature is 35°C. Use F = 1.8C + 32 to convert. Then find the inverse function C(F).

8. Is P(g) one-to-one? Explain.

Part 3: Charging & Brightness

Day 3 · B(t) = 100 − 80e^(−0.04t), t in minutes.

9. Find B(0), B(30), and B(60).

10. Solve B(t) = 90 using logarithms.

11. Brightness has a maximum 900 at 1 p.m. and a minimum 100 at 1 a.m. Find the midline, amplitude, and period.

Final Product: Battery Lab Report

Your final product must include:

- Battery models with graphs
- Piecewise data plan
- Conversion and inverse
- Charging time with logarithms
- Lab report

Reflection

R1. How can you tell from a table whether data is linear or exponential?

R2. Why does the charging model level off at 100?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Models for battery, charging, plans, and brightness are built and used correctly. Goes beyond the minimum with a creative, realistic choice.	Models for battery, charging, plans, and brightness are built and used correctly.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Lab report includes models, graphs, calculations, and conclusions. Polished, organized, and visually engaging.	Lab report includes models, graphs, calculations, and conclusions. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.